AIMS 5710 Deep Learning Fundamentals and Theories
Homework 3

General Guidelines:

- Please check the submission deadline on **Blackboard** and submit your solutions via **Blackboard**.
- Each homework's deadline has a grace period of a week.
- All other late submissions will be given 0 points with no exception.
- **Do not** close your browser of app before you have successfully uploaded your files. It is your own responsibility of keeping your file integrity.
- Round your results to 3 decimal places or keep the fractional numbers.

1. (50 points) **Backpropagation of matrix multiplication.** The key operation in transformer is to calculate the query-key multiplication matrix.

   Forward input: $Q \in \mathbb{R}^{n\times d}$, $K \in \mathbb{R}^{n\times d}$, Forward output: $Z = QK^T$, where $Z \in \mathbb{R}^{n\times n}$

   Backward input:

$$\frac{\partial J}{\partial Z} = \begin{bmatrix} \frac{\partial J}{\partial Z_{11}} & \frac{\partial J}{\partial Z_{12}} & \cdots & \frac{\partial J}{\partial Z_{1n}} \\ \frac{\partial J}{\partial Z_{21}} & \frac{\partial J}{\partial Z_{22}} & \cdots & \frac{\partial J}{\partial Z_{1n}} \\ \vdots & \vdots & \vdots & \vdots \\ \frac{\partial J}{\partial Z_{n1}} & \frac{\partial J}{\partial Z_{n2}} & \cdots & \frac{\partial J}{\partial Z_{nn}} \end{bmatrix}$$

   Find the backward outputs $\frac{\partial J}{\partial Q_{ij}}$ and $\frac{\partial J}{\partial K_{ij}}$ for $i = 1, \ldots, n$, $j = 1, \ldots, d$.

2. (50 points) **Back-propagation of smooth L1 loss.** The smooth L1 loss's forward computation is given as

   Forward inputs: $z^{(1)}, z^{(2)}, \ldots, z^{(N)}$ and ground-truth targets $\tilde{z}^{(1)}, \tilde{z}^{(2)}, \ldots, \tilde{z}^{(N)}$.

   Forward output:

$$J = \frac{1}{N}\sum_{i=1}^{N} \text{SmoothL1}\left(z^{(i)} - \tilde{z}^{(i)}\right),$$

   where

$$\text{SmoothL1}(x) = \begin{cases} 0.5x^2 & |x| < 1, \\ |x| - 0.5 & \text{Otherwise.} \end{cases}$$

   Find the backward output $\dfrac{\partial J}{\partial z^{(i)}}$ for $i = 1, \ldots, N$.